

Introduction

Building a competitive Fantasy Premier League (FPL) squad under a strict £100 m budget requires balancing star power with cost efficiency—an enduring challenge for millions of managers worldwide. Early approaches focused on **exact optimization** of expected point forecasts within FPL’s complex rules (Eilertsen, Gupta, & Kristiansen, 2018), while time-series methods in related sports (Robinson, 2020) highlighted the trade-off between statistical rigor and practical ranking. As the field evolved, **machine learning** techniques (Pokharel et al., 2022; Valouxis, 2023) surpassed simple heuristics in predictive accuracy, and recent work has wrapped these models into **interactive dashboards** to democratize squad selection (Rajesh et al., 2022).

This project builds on that trajectory, combining linear and tree-based regression models with a straightforward “points-per-million” value metric to forecast 2023–24 season totals and identify high-value players. The following review traces the field’s major milestones and debates, setting the stage for our methodology and application.

Literature Review

Early Optimization & Forecasting (2018–2020)

The first rigorous application of optimization in FPL emerged in 2018, when Eilertsen, Gupta, and Kristiansen formulated a **forecast-based mathematical model** that used rolling-horizon heuristics driven by point forecasts (average, regression, and bookmaker odds) and embedded game-chip constraints into an integer program. This thesis established two foundational pillars: (1) the value of integrating **point forecasts** into squad-selection optimization, and (2) the importance of modeling special FPL mechanics (wildcards, bench boost, free hit) within the decision problem.

Shortly thereafter, Robinson (2020) tackled a parallel challenge in **NFL fantasy football**, using ARIMA time series to project season-long fantasy points and rank players algebraically by forecasted haul. Though in a different sport, it highlighted early on the trade-off between statistical rigor (time-series) and practical ranking formulas, foreshadowing later debates over model interpretability versus predictive power in FPL.

Rise of Machine Learning (2022)

By 2022, FPL research pivoted sharply toward **machine-learning regression**. Pokharel et al. demonstrated that **XGBoost** could predict player ROI effectively, achieving a mean RMSE of ~2.05 points and revealing that mid-week fixture congestion had only minor effects on FPL

returns . Their work marked a turning point: moving beyond pure optimization to **data-driven, nonlinear predictors** that better capture complex interactions between fixtures, rotations, and form.

Concurrently, Valouxis (2023) delivered a comprehensive **closed-source ML pipeline** that not only forecasted an “expected value” (EV) metric via tree-based and linear models but also incorporated transfer recommendations and live-rank performance—his team ranked first among 25,000 managers in a GW24 mini-league . This study underscored two emerging patterns: (1) the importance of **open sourcing** for reproducibility, and (2) the integration of **community feedback** (actual league standings) as a real-world benchmark.

Towards Recommendation Dashboards (2022)

Finally, Rajesh et al. (2022) bridged analytics and usability by building a **player-recommendation system** with an interactive dashboard, using both statistical measures and machine-learning classifiers to counteract **favoritism bias** and democratize squad decisions . This work represents the latest trend of turning predictive models into **actionable tools** for everyday managers, emphasizing visualizations and human-centric design.

Key Debates & Patterns

1. **Optimization vs. Prediction:** Early work (Eilertsen et al., 2018) focused on exact mathematical optimization; later studies (Pokharel et al., Valouxis) showed that **ML regressors** often yield lower error at the cost of interpretability.
2. **Interpretability:** Simpler models (linear, tree-based) continue to be favored for actionable insights (Valouxis, 2023), whereas black-box ensembles (XGBoost) excel in accuracy (Pokharel et al., 2022).
3. **Integration of FPL Mechanics:** Only the earliest and most recent works explicitly model game-chips and real league standings, highlighting an ongoing challenge of embedding FPL’s unique rules into analytics.
4. **Usability & Open Science:** The field has shifted toward **public, interactive dashboards** (Rajesh et al., 2022) and open codebases (Valouxis, 2023), reflecting a broader movement toward transparency and community engagement.

References

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